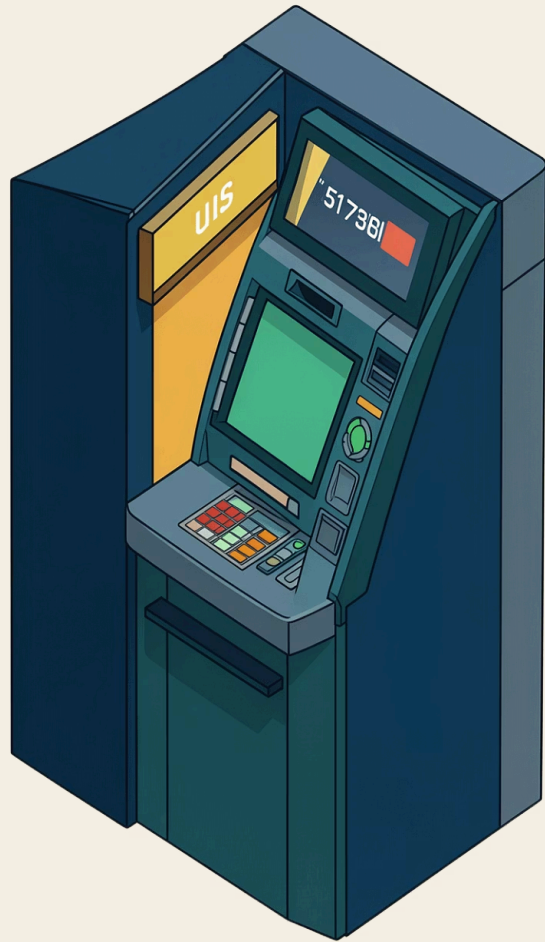




ATM Transaction Simulator

Understanding the Complete Process — A Student's Guide to System Architecture, Security, and Error Handling

What is an ATM Transaction Simulator?

Safe Practice Environment

Software that mimics real ATM behavior, letting users practice banking transactions without risk.

Security Protocol Demonstration

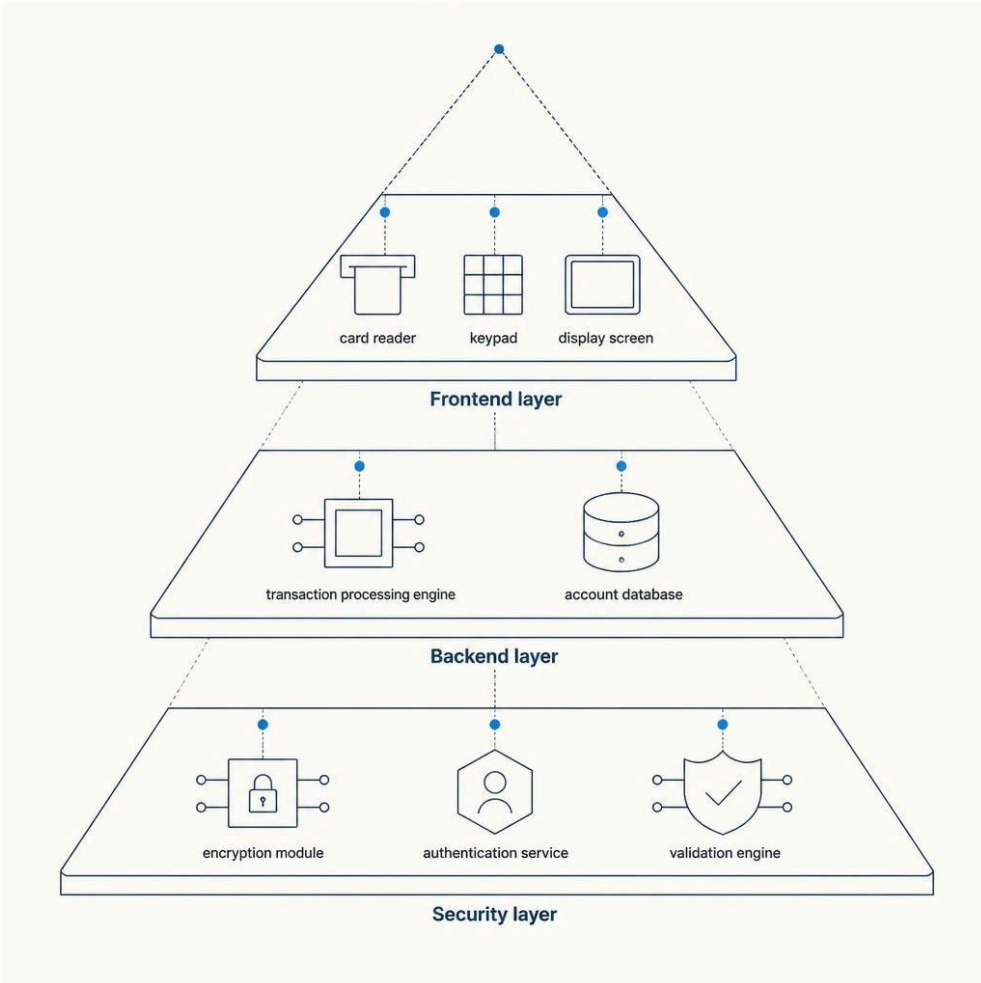
Showcases real-world validation mechanisms, encryption flows, and authentication processes.

Educational Tool

Bridges the gap between theory and practice for students learning financial systems.



System Architecture Overview



Three Core Layers

Frontend

User interface — card reader, keypad, and display screen where customers interact.

Backend

Transaction processing engine and database managing all account data and logic.

Security Layer

Encryption, authentication, and validation protecting every transaction.

Core Components



Card Reader Module

Validates card format and checks expiration date before any session begins.



Account Database

Stores user balances, transaction history, and account metadata reliably.



PIN Verification

Encrypts the entered PIN and matches it against securely stored credentials.



Transaction Engine

Orchestrates withdrawals, deposits, and transfers end-to-end.

Transaction Workflow: Step-by-Step



Insert Card

Validate PIN

Select Type

Check Balance

Each step involves a handshake between the frontend, backend, and security layer — ensuring every transaction is authenticated before it executes.

Security & Validation Features

PIN Encryption

SHA-256 hashing protects credentials — the plaintext PIN is **never stored** anywhere in the system.

Card Validation

The **Luhn algorithm** verifies card number integrity; expiration date is checked before any session proceeds.

Session Management

Auto-logout triggers after a period of inactivity, preventing unauthorized access to an open session.



Error Handling Mechanisms



How the System Recovers Gracefully

→ Invalid PIN

Limits attempts to **3 tries**, then temporarily locks the card to prevent brute-force attacks.

→ Insufficient Funds

Rejects the transaction and displays the current available balance to the user.

→ Network Failure

Queues the transaction for retry or performs a full rollback to maintain data integrity.

→ Card Expired

Blocks the transaction immediately and prompts the user to request a card renewal.

Transaction Types & Limits



Withdrawal

Daily limit **\$500** · Per-transaction maximum **\$300**

Debits account instantly upon successful authentication.



Deposit

Accepts **cash or checks** · Balance updated after verification

Verification delay may apply for check deposits.



Transfer

Between accounts · **Recipient validation required**

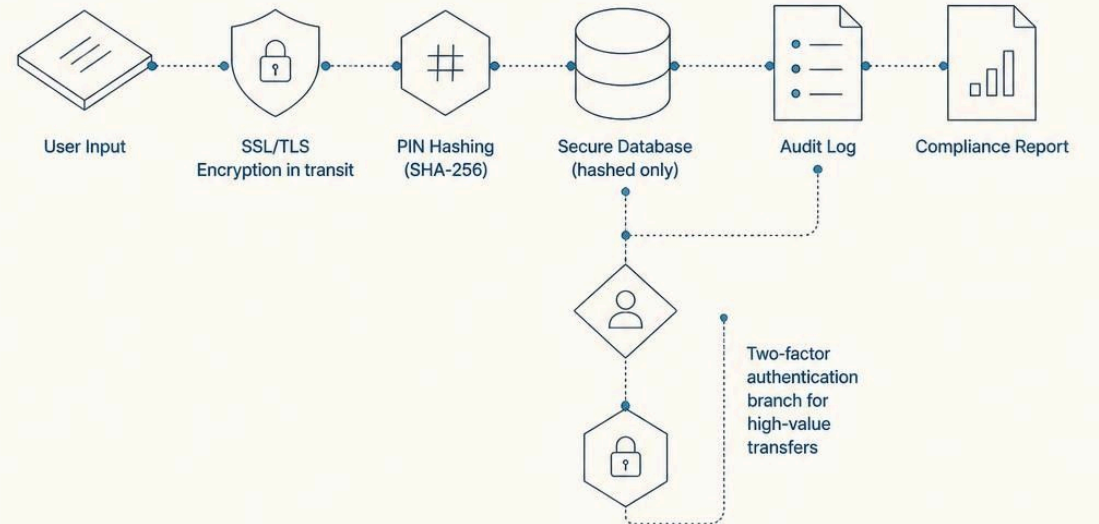
High-value transfers may trigger two-factor authentication.

Data Flow & Security Considerations

End-to-End Data Protection

- **SSL/TLS encryption** secures all sensitive data in transit between client and server.
- The database stores only **hashed PINs** — plaintext credentials are never persisted.
- **Audit logs** record every transaction, ensuring full compliance and traceability.
- **Two-factor authentication** is enforced for high-value or flagged transfers.

☐ Every layer of the system is designed with a "never trust, always verify" security philosophy.





Key Takeaways

Real-World Skills

ATM simulators teach genuine banking security practices applicable to production financial systems.

Multi-Layer Validation

Layered defenses — from Luhn checks to session management — work together to prevent fraud and errors.

Robust Error Handling

Graceful recovery from failures builds user trust and ensures long-term system reliability.